

In a shipyard, a loaded transporter does not drive on the road. It owns the road.

Network-bound capacity, deadlock-free orchestration and fleet-network co-design
(working title)

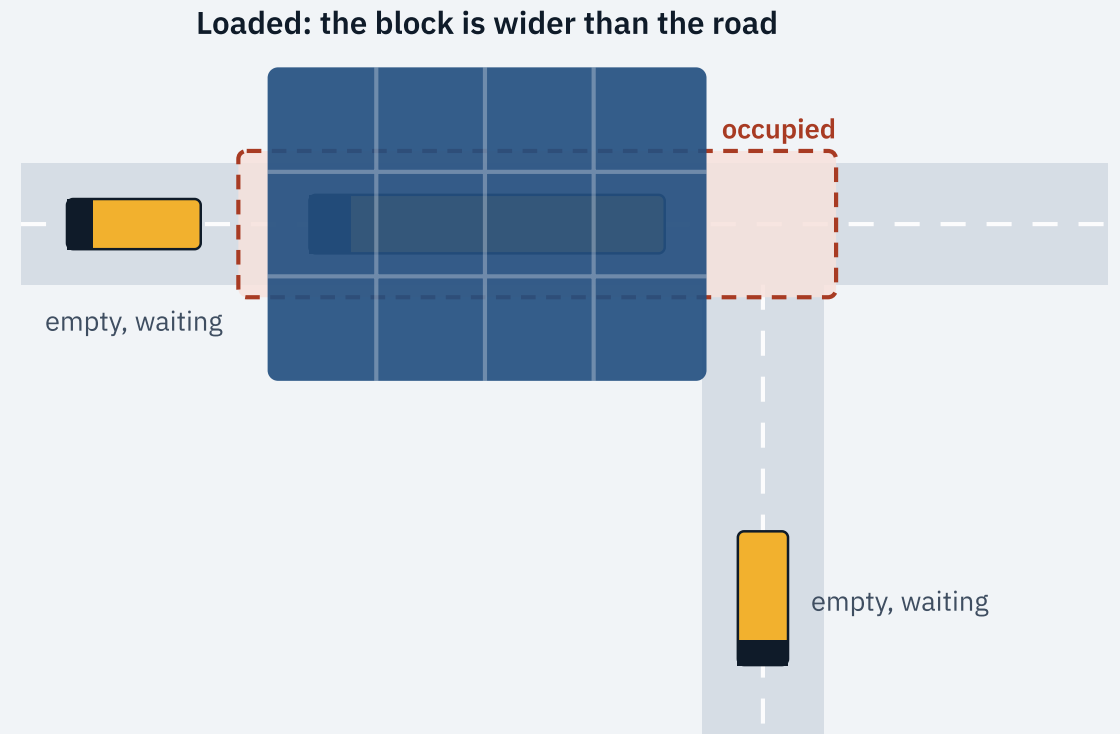
Road-occupying transport in shipyards: network-bound capacity, deadlock-free orchestration and fleet-network co-design

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Manifesto v1 · T22–T24 reviewed (10 Oct 2026) · six sub-studies

Free-flow model: vehicles never meet

Occupancy model: loaded vehicles own the road



The whole thesis in one minute

1

The premise

Shipyard block-transport scheduling almost always assumes a free-flow network: travel times from a distance matrix, vehicles that never interact.

2

The question

When loaded vehicles own the road, what sets capacity, and how should fleet, orchestration and network be designed together?

3

Results so far

A rule-independent bound $T1'$, verified on over 50,000 points in two yards. Okpo's capacity interval is $[627, 1,720]$, its floor set by whole-route reservation. At Okpo's mean of 339 tasks a day it needs 27 vehicles ($< K^* = 50$); at the peak of 678, 117 (16 h).

4

Next

T27: Okpo's new map and deadlock-free rules less conservative than safe release (chapter 4's start); T25: retrofit simulations and the first exchange-rate table; rewriting the $T1'$ and $T1''$ proofs; the first paper.

Capacity is an interval: the network sets its upper edge, orchestration its lower edge.

The order of this briefing (click to jump)

1

Background and premise

Blocks wider than roads; free flow by default; a gap named by peers p. 4

2

Question, claims, hypotheses

Manifesto v1: core question, C1–C5, H1–H7 p. 7

3

Model and theorems

Blocking-time model; theorem list and status p. 11

4

Research map

Six sub-studies, each with its own briefing p. 13

5

Results so far

Bound, capacity interval, work zones, dock mouth, retrofits p. 14

6

Papers and venues

Seven chapters, four papers; venues from the school's 2026 list; graduation requirements p. 17

7

Progress and plan

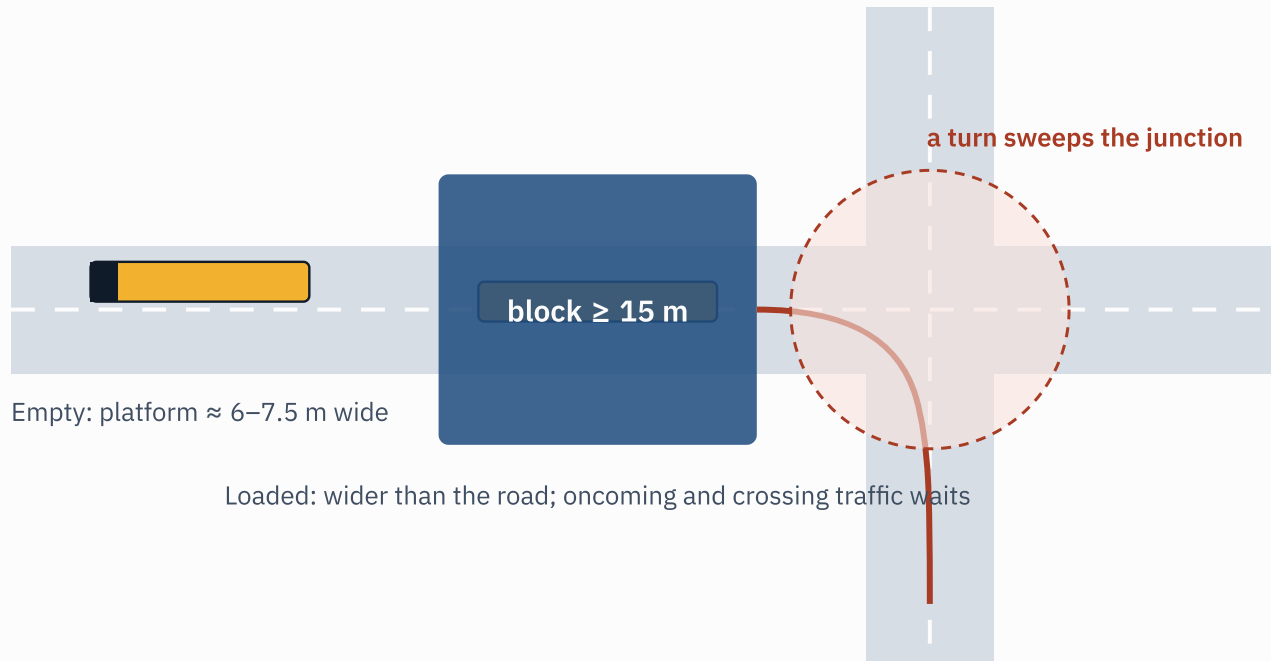
Tasks, eight-week plan, objections, risks and mitigations p. 21

8

Earlier work

How fleet sizing and segmentation serve the thesis p. 26

Blocks are far wider than their carriers: once loaded, a transporter holds the whole segment and junction



- **Empty platforms** are about 6–7.5 m wide; one Hanwha Ocean transporter is 21 m × 7.5 m and drives with signallers ahead and behind
- **Loaded outlines** are often above 15 m, e.g. blocks of about 15 × 15 × 12 m
- **The largest carriers** have platforms of about 30 m × 10 m and occupy the road even when empty

The gap between loaded and empty, and between large and small carriers, is particular to shipyards and weak for AGVs and warehouse robots.

The field's default: a free-flow network in which vehicles never interact

TYPE	WORK	TREATMENT	DISTANCE TO OCCUPANCY
Excluded	Meng et al. 2018	Path overlap in the stockyard only; road interference explicitly excluded	free flow
Free flow	Liu et al. 2022; Zhou et al. 2025; Han et al. 2024	Distance matrices or departure and arrival events only; some recommend fleet sizes on that basis	free flow
Turning delay	Moon et al. 2017	Per-vehicle penalties for turning at junctions, changing each vehicle's shortest path	not occupancy between vehicles
Soft penalty	Roh & Cha 2011; Heo et al. 2013	Interference counted in the objective; penalty when two blocks share a segment at the same time	not a structural capacity constraint; loaded and empty not distinguished; no fleet or network design
Single-yard simulation	Tao et al. 2017	Road capacity and “road interference” in a simulation; utilisation rises after reconfiguring vehicles	mechanism unclear; one yard

No one has treated the exclusive use of segments and junctions by loaded vehicles as a structural capacity constraint, or used it to size fleets or upgrade networks (within our search).

Sources: direction document §1.1, checked against the originals on 9 Oct 2026. Full texts: Meng 2018 (line 302), Liu 2022, Zhou 2025 (assumes “traffic congestion not considered”), Roh & Cha 2011 (eqs. 2, 7), Heo 2013, Moon 2017; Han 2024 is a two-page poster, Tao 2017 abstract only. Search scope: two searches of 8 Oct 2026 and the literature library (shipyard scheduling, multi-agent path finding, railway capacity, queueing networks with blocking).

Peers have named the gap, and the yard data show it

A stated gap: Sun, Liang, Chang (2024), *PLOS ONE*

“how to address the interference limitations of roads in the yard remains a limitation of this study. In actual transportation operations, some roads can only accommodate one transporter at a time.”

Chang Daofang's group, Shanghai Maritime University; they may well follow up, so speed matters.

Yard evidence: Wang, Yin, Khan (2024), *JMSE*

288 delays at a Shanghai yard in 2022: transporters take several lanes and other vehicles must give way; the paper assigns each road-environment event (other vehicles, obstacles, pedestrians) a delay of about 15 min (an assigned value, not a calibration anchor).

Other fields: more vehicles jam, road changes move capacity

- **Container-terminal AGV simulation:** going from 60 to 80 vehicles lowered hourly throughput (by about 2%) Cheng et al. 2005
- **Terminal traffic network:** saturation near 100 vehicles; without keep-clear hatching at junctions, deadlocks from 30 Zhou et al. 2017
- **Underground mine:** single-width roadways with passing bays, trucks managed on railway block-section logic Eustace et al. 2021

These are single-site simulations or operating observations; the shipyard's loaded-to-empty footprint ratio is the most extreme of them.

The core question: when loaded vehicles own the road, what sets capacity?

When loaded vehicles occupy the road exclusively, what determines the capacity of a shipyard block-transport system, and how should fleet, orchestration and network be designed together?

The usual view

- Transport capacity = fleet capacity
- Scheduling means assigning jobs to vehicles
- More vehicles are better, with diminishing returns at worst

The road-occupancy view

- Loaded vehicles hold segments and junctions while passing; loaded and empty occupancy are asymmetric
- Capacity is an interval: the network sets the upper edge, orchestration the lower
- Where the network binds, dispatching becomes road scheduling; where the bound is tight, road changes can replace vehicles

What sets capacity?

bounds and interval · ch. 3



How to use it fully?

deadlock-free orchestration · ch. 4



How to expand it?

vehicles or roads · ch. 5



How far does it carry?

cross-yard prediction · ch. 6

Five central claims in one figure

C1 Two upper bounds and the capacity interval

Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound follows from the blocking-time structure and is computable by linear programming. Attainable capacity lies in the interval [best deadlock-free rule, network bound]

C2 Fleet-throughput trapezoid

Throughput first rises linearly with fleet size, then reaches a network-set plateau; when waiting occupies roads it declines past a jamming threshold. Free-flow models see neither of the last two

C3 Dispatching becomes road scheduling

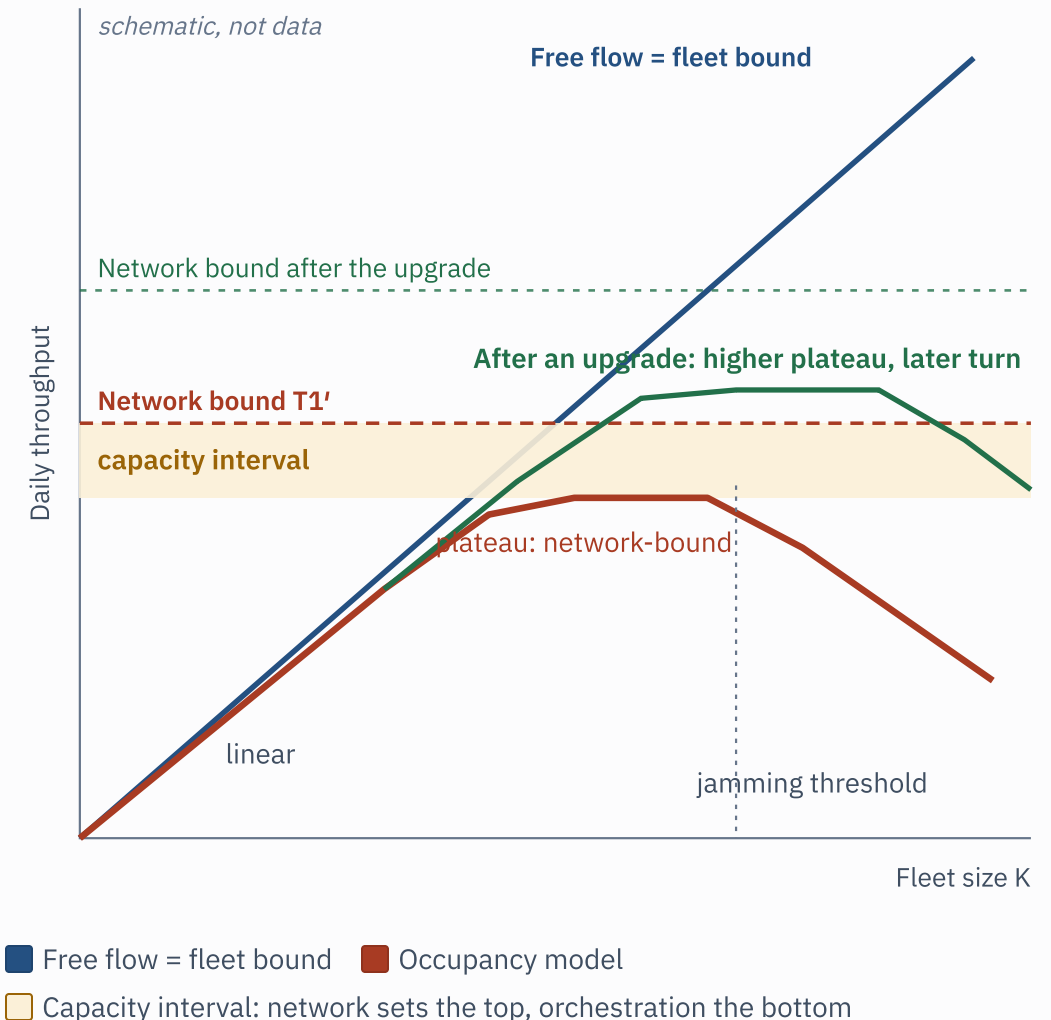
Where the network binds, how much of the bound is attained is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks

C4 Road-vehicle exchange rate

Where the bound is tight, upgrading key elements can replace vehicles, at the dual price converted into vehicles. When the plateau sits far below the bound, change orchestration first. This is the boundary between C3 and C4

C5 Transferable

The network bound computed straight from the map, and its binding elements, predict plateau, jamming threshold and upgrade value across yards



Seven hypotheses, each conclusive either way (1): H1-H3

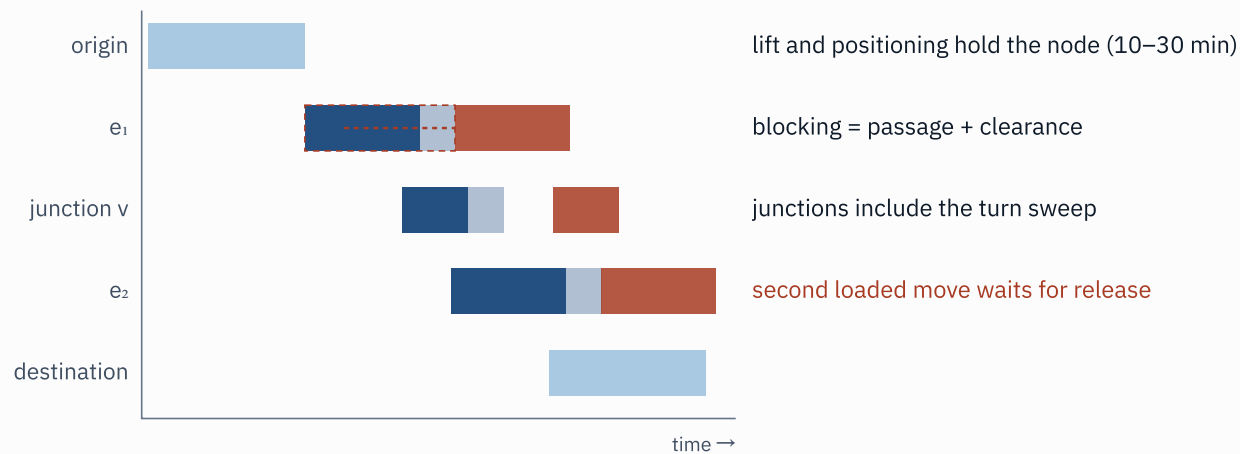
#	HYPOTHESIS	CRITERION	HOLDS / FAILS
H1	Network-bound regime: at realistic daily volumes and fleet sizes, the system is network-bound, at least at erection peaks or near the dock mouth	$K^*(20\%)$: marginal gain first and permanently below 20% of free flow; vehicles needed $\geq K^*$ means network-bound; plus “daily volume / network bound”; whole-yard volumes, one conversion for both yards	Network-bound capacity exists / a density criterion for where free flow suffices
H2	When the decline appears: with off-road waiting and deadlock-free rules throughput never falls (a plateau); when extra vehicles occupy network resources (waiting on roads, incl. for cranes; finite holding area; physical or no deadlock clearing) it falls beyond K_0 , which moves right with holding capacity S ; good orchestration delays K_0	Read saturated curves under no-teleport rules; scan holding capacity S , clearing mode and where vehicles wait for cranes, one at a time	Conditions under which vehicles hurt, and what a holding area is worth in vehicles / a redundancy index (marginal gain vs free flow)
H3	Free-flow error: free flow underestimates completion time and vehicles needed, superlinearly in vehicle density	Service-time gap $\Delta(K) \propto K^\alpha$, $\alpha > 1$; the ratio of vehicles needed read at whole-yard volumes	Systematically wrong at peaks / error under control, with a correction factor

Source: research manifesto v1 §3. H1 is reported in three parts: the capacity interval and its start K^* ; vehicles needed at whole-yard volumes relative to K^* and to free flow; whether the dock mouth caps the yard, which depends on whether vehicles waiting for cranes block the way to other stops.

Seven hypotheses, each conclusive either way (2): H4–H7

#	HYPOTHESIS	CRITERION	HOLDS / FAILS
H4	Road–vehicle exchange rate: where the network binds, one upgrade of a key element is worth at least one vehicle, and dual prices identify the element. Options: segmented blocking, dock-front holding bays, passing bays, one-way loops, widening, a second entrance	Simulated change in vehicles needed vs. the dual-price prediction; agreement is expected only where plateau / bound is near 1	The exchange rate holds / the bottleneck is orchestration; chapter 4 is strengthened
H5	Safe orchestration: naive dispatching deadlocks or nearly so at peaks; a provably safe policy loses little throughput	Deadlocks counted under no-teleport rules; safety cost = loss of the best safe rule relative to the network bound	Safe orchestration is cheap / a cost formula for safety and its conditions
H6	Cross-yard prediction: the network bound computed from the map, with its binding elements, predicts plateau, K^* and upgrade value across yards; topology explains why the bound binds where it does	Leave-one-out; classes by key element: dock-front single lane, door type ...	A two-second LP screens a new yard / what transfers is the bound tool and the key-element classes
H7	Vehicle type and occupancy: large units occupy loaded and empty; small ones yield when empty but run extra empty trips to team up. Case: multi-vehicle moves of giant blocks to floating docks	Compare the best fleet mix across degrees of network constraint	The best fleet mix shifts with network constraint / vehicle type and network can be decided separately

The model: segments and junctions are exclusive resources held for a blocking time



- **Road graph** $G = (V, E)$: segments e and junctions v are exclusive resources
- **A loaded move** holds each resource on its path for a blocking time, as in railway signalling
- **Empty runs by class**: large units occupy even when empty; small ones can pass on passable links
- **No road widths measured**: passability classes (loaded-exclusive / passable by empties / two-way) are scenario parameters; results are functions of the passable share
- **Resource granularity**: long links can be split at their stops with direction locks; the granularity follows the imagery check of the binding elements

One move holds: $\tau_r = \text{pass}_r + \text{clear}_r, r \in \text{path}$

The theorem list and where each stands

NO.	STATEMENT	STATUS
T1'	The network part of the two bounds: an occupancy-rate LP with width bin-packing rows, crane rows and window-edge terms. Holds for rules satisfying (A1)–(A3); loaded routes fixed	first draft; author rewrites
T1''	Route-free bound: loads also as multicommodity flow; holds for any routing rule	draft (T23); author rewrites
T2	Trapezoid law: on corridors and loops, piecewise throughput in K and the jamming threshold under asymmetric loaded/empty occupancy	to do
T3	Unbounded free-flow error: a constructive proof	author
T4	Non-monotonicity: a single-lane corridor with a holding area of capacity S ; under greedy release throughput strictly falls for $K > K_0(S)$	to do
T5	Safe orchestration: reserving to the next safe holding point cannot deadlock, with a loss bound relative to the upper bound	T27 testing
T6	Road–vehicle exchange rate from dual prices; only upgrades on tight constraints pay. New links use T1''	author
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from blocking job shop)	to do
Candidates	Dock-mouth proposition: with nominal load $\rho < 1$, blocking-induced starvation destabilises the system; off-road waiting or a segmented dock road restores stability. The other is attainability of T1'	T22 supports; author writes

Source: research manifesto v1 §4 (statements and status). Highlighted: the theoretical core of the first paper. Proofs and text are written by the author.

Research map: one thesis, six sub-studies, each with its own briefing

CH. 2–3

Occupancy model and network bounds

With loads owning the road, what is the rule-free capacity bound?

C1 · T1' · T1''

T1' verified on two yards; system T1' and route-free T1'' done (T23)

[Open →](#)

CH. 3

Capacity interval and the network-bound regime

Which interval holds attainable capacity, and when does the network bind?

C1 · C2 · H1 · H3

Capacity interval and work zones done (T22); H1's main caliber is the author's call

[Open →](#)

CH. 3

The dock mouth at erection peaks

Where do vehicles wait at erection peaks, and can the dock mouth hold up the yard?

H1 · H2 · dock proposition

Holds without teleporting; threshold moves left with fleet size (T22)

[Open →](#)

CH. 4

Traffic orchestration

How to order passages without deadlock and push the lower edge up?

C3 · H5 · T5

Safe release done (T22); less conservative rules in T27

[Open →](#)

CH. 5

Fleet–network co-design

Buy vehicles or change roads? How many vehicles is an upgrade worth?

C4 · H4 · H7 · T6

Two examples of the C3 / C4 boundary (T22); exchange-rate table in T25

[Open →](#)

CH. 6

Testing across shipyards

Does the bound computed from a map predict across yards?

C5 · H6

Two yards done; maps and base maps ready for 20 yards; group A being traced

[Open →](#)

Ch. 2 defines the object



Ch. 3 where capacity comes from



Ch. 4 using it fully



Ch. 5 spending to expand it



Ch. 6 which yards it reaches

Sources: direction document §4 (chapter plan); research manifesto v1; task board T17–T27 and yard list (10 Oct 2026). Click a card to open its sub-study briefing.

Results so far: both edges of the interval stand, and the dock-mouth effect holds without teleporting

51,767 / 0

The bound holds

None of the 51,767 distinct T21 and T22 occupancy points exceeds its T1'. System T1' (no sampling noise): Okpo 1,736, Yantai 1,298; the route-free T1'' holds for any routing rule.

[627, 1,720]

Capacity is an interval; reservation sets the floor

Among the deadlock-free rules tested, whole-route reservation is best; the new safe release is exact and never deadlocks but is conservative, below reservation for $K \geq 20$. In both yards the floor reaches about 36% of the ceiling.

27 / 117

Okpo itself: outside the regime on average, inside at peaks

At Okpo's daily mean of 339 tasks, whole-route reservation needs 27 vehicles, below $K^* = 50$; at the peak of 678 it needs 117 (16-h day). On a 24-h day: 15 and 46.

where to wait

Dock mouth: drops without teleporting too

Rules that wait for the crane on the dock road, including the non-teleporting safe release, all fall as crane load p rises, with the threshold moving left as fleets grow; reservation, which waits off-road, does not fall. Splitting the dock road at its

+76% / +8%

The C3 / C4 boundary has examples

In the erection mix, where the dock binds, segmenting lifts T1' by 76% and the reservation plateau by 8.3%; in the main scenario it lifts only the ceiling (+8%) and the plateau barely moves (+0.3%): the first rewards road changes, the second orchestration.

both sides

The two yards fall on either side of H1

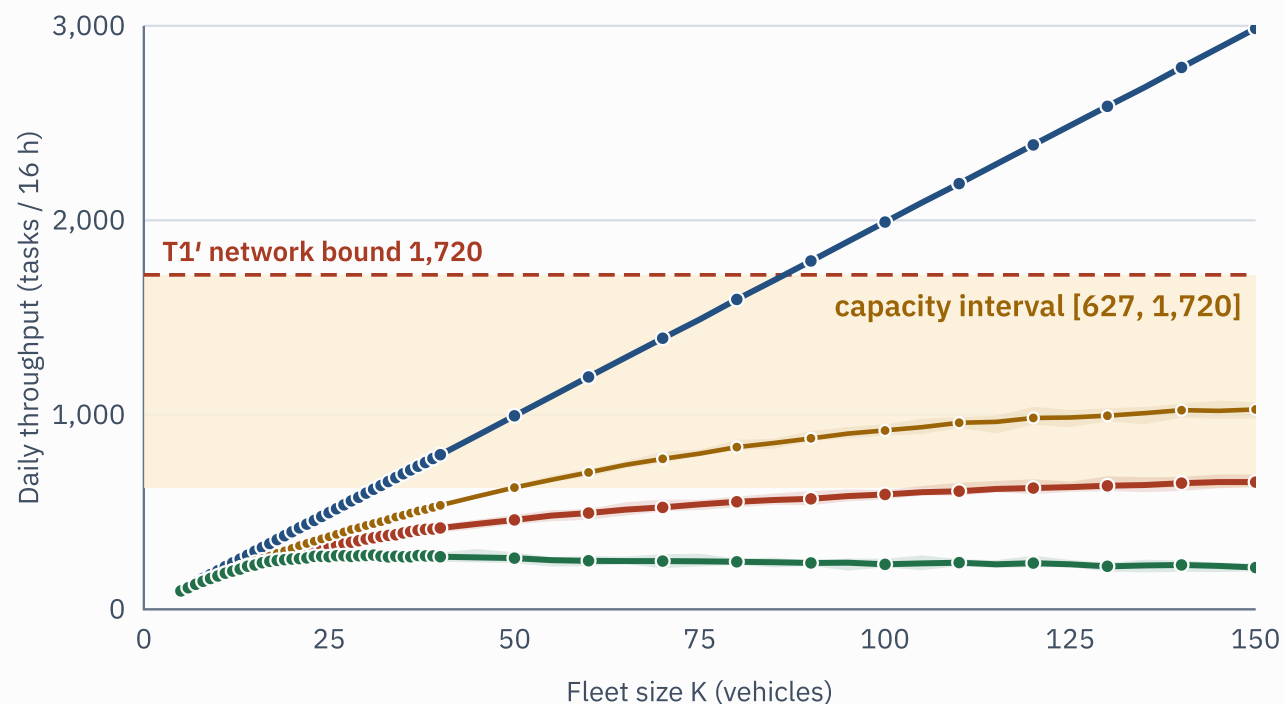
Yantai's own 23–46 tasks a day need 1–2 vehicles; Okpo enters the regime at peaks. Key elements come in two types, the dock-front single lane (road 161) and the door (building 014), both checked by the author on imagery.

Sources: the review (not reviewed): T22 report §0, 3, 4, 6, 7; T23 report §0; reviews T22, T23, T24; task board, "author's imagery check". Conditions: MY-B task mix, Okpo map rev 1825, Yantai r396, dock road as one resource (segmenting noted), saturated or finite demand as per section; segment request (reference rule, teleport clearing) supports no conclusion on its own. H1's main caliber (16 h / 24 h, mean / peak) is the author's decision after T27.

The signature figure: free flow keeps rising, occupancy capacity stays inside the interval

Okpo

Yantai



Free flow Whole-route reservation Safe release Segment request (reference)

capacity interval

627-1,720

Capacity interval (a day): lower edge the whole-route reservation plateau (mean of $K = 100-150$), upper edge T1'

36%

The lower edge reaches only this share of the upper: orchestration decides the rest, which chapter 4 aims to win back

$K \approx 50$

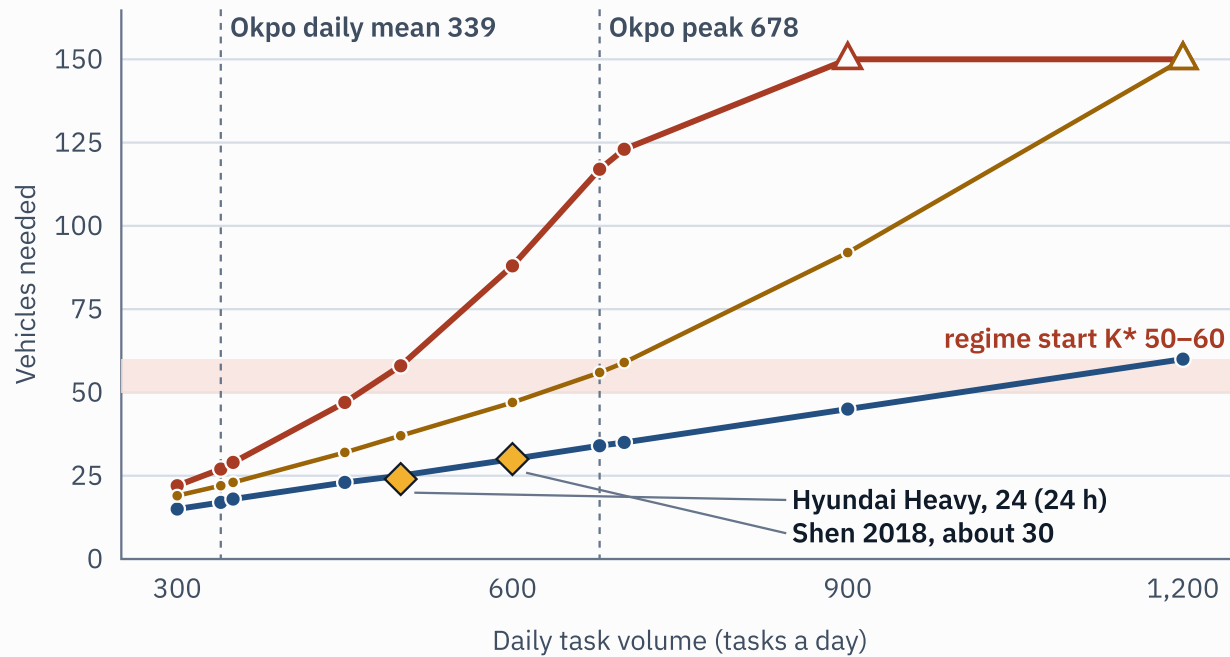
Network-bound from here (marginal ratio first below 20%; range 50-60)

Sources: thesis-core a882742; Okpo results/a0ab16fceb94, 49d2383872e7 and safe release 656e2da75368, Yantai b82aa19b22af and 930e637e0658; saturated demand, means of 10 seeds, shading min-max. Throughput = transport tasks per 16-h day. T1' bounds only the occupancy models; system T1' (no sampling noise, T23): Okpo 1,736, Yantai 1,298. Segment request is a reference rule (teleport clearing, 900 s) and supports no conclusion on its own.

Work zones: vehicles needed rise with daily volume, and field fleets sit between free flow and reservation

16-h day

24-h day



■ Free flow
 ■ Whole-route reservation
 — Segment request (reference)
 ■ Field fleets

regime start K^* 50–60

27

Okpo daily mean of 339: vehicles needed under whole-route reservation, below $K^* = 50$ (16-h day)

117

Okpo peak of 678: vehicles needed under whole-route reservation (16-h day)

30 → 88

600 a day (large Korean yard scale): free flow → whole-route reservation (16-h day)

Sources: thesis-core a882742, T22 report §6.1 and 7 (results/t22_report_3287d62c7916/t22_demand.csv): Okpo main scenario, dock road as one resource; vehicles needed = smallest integer K whose 10-seed mean throughput reaches $0.99 \times$ the daily volume; open triangles = not reached at $K = 150$. Okpo's 339 a day is the same-method conversion (294–363 across years, 3% from Hanwha's direct anchor of 350); peak = $\times 2$; Hyundai Heavy about 500 moves, 24 vehicles, 24 h (Park & Seo 2006); Shen et al. 2018 about 600 a day, about 30 vehicles (shifts not stated) (T24, full texts). K^* interval 50–60 from T18. The main caliber is the author's decision after T27.

Thesis plan: seven chapters and four journal papers answering one question

CH.	TITLE	QUESTION	OUTCOME	CLAIMS · HYP.	PAPER
1	Introduction	Where free flow comes from and why it fails in yards	A literature map across four fields	—	—
2	Occupancy model	How to represent wider loads, lifting at nodes, swept junctions	Blocking-time graph; complexity; public-data instance generator	C1	Paper 1
3	Capacity	Where capacity comes from, when the network binds, when the dock mouth destabilises	T1', T1'', T2–T4; capacity interval and work-zone chart	C1 · C2 · H1–H3	
4	Orchestration	How to assign jobs and order passages without deadlock	Exact model, large-scale search, T5, rolling online	C3 · H5	Paper 2
5	Fleet–network co-design	Buy vehicles or fix roads; large or small vehicles	T6; exchange-rate table; control granularity; bilevel algorithm	C4 · H4 · H7	Paper 3
6	Many yards	Do the laws generalise and predict?	15 public layouts; key-element types; leave-one-out prediction	C5 · H6	Paper 4
7	Conclusion	—	—	—	—

Sources: direction document §4 (chapter plan); research manifesto v1 (claims and hypotheses); the author's review and guidance §4.2–4.3 (outputs of chapters 3 and 6). Highlighted: the theoretical core of the first paper.

Where the four papers go: all from the school's 2026 journal list, B-class first, A-class as stretch goals

PAPER	FIRST CHOICE (B)	STRETCH (A) AND ITS CONDITION	ALTERNATIVES (B)	SUPPLEMENTARY (C)
Paper 1 Ch. 2–3 · bound, interval, dock mouth	<i>Transportmetrica A</i> <i>Ocean Engineering</i>	<i>RESS</i> if the dock-mouth proposition becomes a proved result on blocking-induced instability and recovery	<i>IEEE T-ASE</i> <i>Applied Mathematical Modelling</i>	<i>CAIE · JSPD</i>
Paper 2 Ch. 4 · orchestration	<i>IEEE T-ASE</i> <i>Transportmetrica A</i>	<i>IEEE TII · IEEE T-RO</i> rolling online orchestration closed through a digital twin; or provably safe coordination of agents whose footprint grows when loaded	<i>EAAI · ESWA</i> <i>IEEE RA-L</i>	<i>CAIE · AAMAS</i> <i>JMS</i>
Paper 3 Ch. 5 · fleet–network co-design	<i>Ocean Engineering</i> <i>IEEE T-ASE</i>	<i>RESS</i> if written as capacity loss under key-link failure and redundancy design	<i>Transportmetrica A · ESWA</i> <i>ASC · JMP</i>	<i>CAIE · JSPD</i>
Paper 4 Ch. 6 · many yards	<i>Ocean Engineering</i> <i>J. Marine Science and Application</i>	— no natural A-class fit; not forced	<i>EAAI</i> when linked to automatic layout extraction by segmentation	<i>JSPD · IJNAOE</i> <i>JMSE · ISP</i>

Source: the school of shipbuilding's 2026 journal classification (graduation-requirements file, 23 Sep 2026 version, §1 and 7). Transportmetrica A = Transportmetrica A: Transport Science; RESS = Reliability Engineering & System Safety; T-ASE / TII / T-RO / RA-L = IEEE Trans. Automation Science and Engineering / Trans. Industrial Informatics / Trans. Robotics / Robotics and Automation Letters; EAAI = Engineering Applications of Artificial Intelligence; ESWA = Expert Systems with Applications; ASC = Applied Soft Computing; JMP = J. Manufacturing Processes; CAIE = Computers & Industrial Engineering; JSPD = J. Ship Production and Design; AAMAS = Autonomous Agents and Multi-Agent Systems; JMS = J. Manufacturing Systems; IJNAOE = Int. J. Naval Architecture and Ocean Engineering; JMSE = J. Marine Science and Engineering; ISP = International Shipbuilding Progress.

Outputs and graduation requirements: one B-class paper is published, and each thesis paper aims at B-class first

Applying at normal duration
Mid-term evaluation good or better and at least one B-or-above journal paper; then one of: 1 A; 2 B; 1 B + 2 C.

Applying early
Mid-term evaluation excellent and at least one B-or-above journal paper; then one of: 1 A + 1 B; 3 B.

OUTPUT	JOURNAL (CLASS)	STATUS
Segmentation (MY-A)	<i>Ocean Engineering</i> (B)	published 368 (2026)
Fleet sizing (MY-B)	<i>Ocean Engineering</i> (B)	target journal
Paper 1 (ch. 2–3)	<i>Transportmetrica A / OE</i> (B) ; <i>RESS</i> (A)	writing, wk 5–8
Paper 2 (ch. 4)	<i>IEEE T-ASE</i> (B) ; <i>TII / T-RO</i> (A)	from T27
Paper 3 (ch. 5)	<i>OE / IEEE T-ASE</i> (B) ; <i>RESS</i> (A)	from T25
Paper 4 (ch. 6)	<i>OE / JMSA</i> (B)	group A being traced

Source: the school of shipbuilding's PhD graduation requirements (2026 journal list update, 23 Sep 2026; output-count rules carried over from the earlier file, superseded by any new school rules) §0, 2, 3; MY-A in the segmentation briefing, MY-B in the block-transporter-choice briefing. JMSA = Journal of Marine Science and Application; other abbreviations on the previous slide.

Algorithms, and what the thesis brings back to three research communities

Algorithms (chapters 4–5)

- **Exact model:** CP-SAT interval variables, one no-overlap constraint per segment and junction; ground truth on small instances
- **Large scale:** integrated assignment-and-sequencing neighbourhood search; priority-based planning for conflicts; compared with “schedule in free flow, then repair conflicts”
- **Safety layer:** the T5 reservation policy as a shell around any heuristic
- **Rolling online:** re-plan under uncertain completion times; release with forecasts
- **Co-design:** outer level picks upgrades and fleet, inner level evaluates by orchestration; T6 dual prices rank the outer options
- **Optional frontier:** LLM-evolved orchestration rules (EoH, ReEvo, FunSearch; no shipyard use found)

Framing: “we extend X to Y and find that Z changes under Y”

Multi-agent path finding

A new agent class: footprint multiplies when loaded, long lifts at endpoints, fleet size to be chosen; existing large agents have fixed shapes and no tasks

Railway capacity

Blocking times and compression carried to trackless road networks with asymmetric loaded and empty occupancy

Queueing networks with blocking

A real system that breaks the vehicle–hole symmetry, because empty and loaded vehicles occupy differently

Progress: the three first-order corrections passed review; chapter 4's first rule comparison is running

WHEN	ITEM	OUTPUT	STATUS
10-08	T17–T21 flag experiments on two yards; T1 tightened to T1'; erection at crane cadence	Flag curves; dock-mouth mechanism	done
10-09	Author's full review; manifesto v1 frozen; imagery check of road 161 and building 014	Three first-order corrections	done
T24	Literature anchors: Okpo 339 tasks a day; Hyundai Heavy 24 vehicles / 24 h; Tao 2017 full text compared	External comparison; introduction	passed
T22	Safe release rule; segmented dock road; whole-yard volumes (integer K, 16 h / 24 h)	Capacity interval; work zones; dock thresholds by K	passed
T23	System T1'; route-free T1"; route slack; dual-price table	Bound figures; exchange-rate input	passed
T27	Step zero: Okpo rev 1833; less conservative deadlock-free rules: step-wise feasibility search; claims cut at the next safe holding point (T5)	Re-check of floor, H2 decline, H5	running
T25	Retrofit simulations (on rev 1833); door granularity; three-resource dock network	First exchange-rate table	after T27
T26	Floating-dock scenario; finite holding area S	H7 case; K _o (S) for H2	after T25
Author	H1 part 2 numbers, part 3 two-layer wording, main caliber (16 h / 24 h, mean / peak)	Final H1 wording	after T27

Sources: task board T17–T27 (10 Oct 2026); reviews T22, T23, T24 (9–10 Oct 2026); thesis-core a882742.

Eight-week plan: the two weeks of corrections are done; the first paper goes out within 3–4 months

WHEN	THE AUTHOR	COMPUTATION	OUTPUT (PLACE IN THE THESIS)
Weeks 1–2	Manifesto ✓; imagery checks ✓; group-A tracing	T22 ✓, T23 ✓, T24 ✓; T27 (with Okpo's new map) running	Ch. 3 interval, work zones, dock section; bound figures
Weeks 3–4	Rewrite the T1' and T1'' proofs; settle H1's three items after T27; write the dock-mouth proposition (three-resource network)	T25: retrofits, exchange-rate table; T26 floating docks	First exchange-rate table; ch. 5 prototype; final Okpo map
Weeks 5–8	Write the first paper	External comparison table (normalised by cycle time); T1' and plateaus for the first group-A yards	Paper 1 draft (Transportmetrica A or OE); first ch. 6 data

The first paper's four results: (1) model and T1' (system T1', route-free T1''); (2) capacity interval and work zones; (3) the dock mouth: where vehicles wait and how the dock road is segmented, holding without teleporting; (4) the rule separating C3 from C4 (T22's two examples) and the first exchange-rate table.

Sources: the author's review and guidance §4.2 (first paper) and §4.4 (eight weeks); task board T22–T27 (10 Oct 2026; the author set T27 before T25). “Submit within 3–4 months”: direction document §12.

The likeliest objections, and the answers

OBJECTION	ANSWER
Yard conflicts are done already Roh & Cha 2011; Heo 2013; Moon 2017; Tao 2017	They treat crossings as soft penalties in path search or turn time for one vehicle; Tao et al. 2017 (full text checked) is a single-yard simulation with road capacity counted in vehicles. None formalises blocking times, compares rules or handles deadlock, gives bounds, or designs fleet and network.
MAPF already has large agents Li et al. 2019	Their shapes are fixed and they carry no pickup-and-delivery tasks; none combines a load-dependent footprint, long lifts and fleet sizing. The thesis uses them as a source to build on.
Yards have too few vehicles to jam	That is what H1 tests, and T22 tested it cell by cell: at Okpo's own daily mean of 339 the yard is outside the regime under reservation (27 vehicles < 50), at its peak of 678 it is inside (117, 16-h day); at a large yard's 600 a day it holds on a 16-h day too. Either way there is a result.
The dock-mouth drop is a teleport artifact	T22 tested it: the non-teleporting, deadlock-free safe release falls with p too, its threshold moving left as fleets grow; reservation, waiting off-road, does not fall; segmenting the dock road moves thresholds right. The result is stated as a conditional reading of where vehicles wait and how the road is segmented, and as a design lever.
No measured data	The theorems need no data; Okpo's daily volume, converted from public data, is 339 (3% from Hanwha's direct anchor of 350); Hyundai Heavy's 24 vehicles (24 h) and Shen et al.'s about 30 both sit between the model's free flow and reservation on a 24-h day.
Exclusivity is too strong	Exclusivity is a parameter: results are functions of the passable share; resource granularity (whole segment, split at stops, doors) is a scenario parameter too, chosen by the imagery check.

Risks and mitigations

RISK	MITIGATION
Working-day length and the mean / peak caliber change H1's reading (Okpo's peak holds at 16 h, sits near the start at 24 h)	Report both calipers and state under which each holds; the author sets the main caliber after T27. Hyundai Heavy states 24 h and can serve as external evidence.
Safe release is conservative (about 2/3 of its waits come from criterion rejection), so the floor and H2's decline may change with the rule	T27 re-checks with less conservative deadlock-free rules; the floor is always the best rule tested, named, so the interval can only narrow.
Okpo's numbers will shift with map rev 1833 (floating docks, branch roads)	T27's step zero switches the map and compares item by item with rev 1825; every result names its map version, and the paper's figures are redrawn on the final map.
The Chang Daofang group (Sun 2024) follows up	No 2025–2026 follow-up found; the first paper goes out within 3–4 months as planned, with a preprint.
Only two yards so far	All 20 yards on the list now have maps and base maps; group A's six get T1' first (fast), simulations follow; the imagery check of key elements is a fixed step for every yard.

Sources: the author's review and guidance §5; T22 report §2.1, 7, 10 (thesis-core a882742); task board T25, T27 and “author's imagery check”; yard list (10 Oct 2026).

The results ladder: every chapter's floor is a B-class paper on its own; the ceiling aims at A-class

	FLOOR (DONE ONCE DONE)	TARGET	CEILING
3 Capacity	Two bounds and the trapezoid on one yard two yards in hand	T1', T1'', T2–T4 proved; capacity interval and work-zone criterion	Attainability of T1' and the dock-mouth proposition
4 Orchestration	Exact model and large-scale search beating “free-flow schedule, then repair”	T5 safe policy with small loss; the lower edge pushed to yard per-vehicle productivity	LLM-evolved rules beating hand-written ones and transferring across yards
5 Co-design	Grid evaluation of fleets and upgrades	T6 exchange rate and key elements (incl. control granularity)	Bilevel co-design with optimality gaps
6 Many yards	8 yards	15+ yards; key-element types and leave-one-out prediction	A simple formula for jamming thresholds from topology

Earlier work serves as tools; the thesis starts from a new premise

ASSET	USE IN THE THESIS
Fleet sizing (MY-B)	ALNS and CP-SAT code start chapter 4; price data feed chapter 5's vehicle types and costs
Coupled-carriage ideas	Chapter 5, H7: large carriers against teams under occupancy
Centreline networks and the tracing tool	Production tool for chapter 6's layouts
Remote-sensing segmentation (MY-A, published)	Optional: faster layout extraction for many yards

Open the fleet-sizing (MY-B) briefing →

Open the segmentation (MY-A) briefing →

 **In a shipyard, a loaded transporter does not drive on the road; it owns the road. Capacity is capped by the network, not the fleet.**

Thank you.

Source: direction document §9 (use of earlier assets). The thesis starts from the premise of §1; earlier work is used as tools and cited.

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